

1	NUCLEAR RADIATIONS, RADIATION LEAKAGE, AND SHIELDING	263
1	A Radiation Leakage, 263	
1	B Attenuation and Absorption, 274	
1	C Shield-design Considerations, 300	
5	MATERIALS: REQUIREMENTS, PROPERTIES, AND PROBLEMS	331
	A Reactor Materials, 331	
	B Propellants, 361	
	C Thermal Structural Effects, 377	
	D Radiation Damage, 389	
7	REACTOR-DESIGN AND ENGINE-CYCLE CONSIDERATIONS	393
	A Reactor Configurations and General Design, 393	
	B Detailed Technical Aspects of Propulsion-reactor Design, 401	
	C Engine Cycles and Selection, 426	
	APPENDIX A Bessel Functions, 435	
	APPENDIX B Multigroup Reactor Calculations, 436	
	APPENDIX C Graph of First Exponential Function $E_1(x) = \int_x^\infty \frac{e^{-y}}{y} dy$, 440	
	APPENDIX D Conversion Factors and Physical Constants, 441	
	Index 445	

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CHAPTER 1

NUCLEAR FLIGHT PROPULSION

A Historical Background and Introduction

Nuclear reactors for central-station power production are a sound and growing part of our modern technical world. This present globe-wide revolution in power-generation methods for a power-hungry mankind is only the beginning of a nuclear-fueled invasion of heat-power machinery uses which will eventually embrace the propulsion-heat-engine field, from merchant-ship drives to interstellar flight. This revolution will not be completed overnight, just as steam did not replace animal and hydro power for nearly a century; after all, it has taken two decades from the first controllable fission reactor to the central-station power mentioned above. The rate of advance of nuclear energy into new heat-power areas will depend on the amount of support given to engineering development of new nuclear devices and on the skill with which this is carried out. Support can be obtained and held only if the remarkable possibilities inherent in the replacement of chemical by nuclear energy are clearly understood, and their successful achievement is possible only if the engineering problems of nuclear power systems are equally well understood and solved in practical and reliable ways.

To illustrate these points it is instructive to examine the history of nuclear-flight-propulsion programs in the United States. The use of nuclear-fission energy for aircraft and rocket propulsion was proposed and discussed almost from the time of achievement of the first fission chain reactor, in 1942, by Enrico Fermi and his associates in the wartime Manhattan District project. In the later and less hectic days of the Project's effort to develop the atomic bomb, time was found by many people to speculate on the future nonbomb applications of nuclear energy. The possibility of using a fission "pile" to heat a working fluid to high temperatures for use as rocket propellant was discussed briefly as early as 1944 at both the Los Alamos Scientific Laboratory and the University of Chicago Metallurgical Laboratory. Outside the laboratories one of the first unclassified publications on rocket propulsion was a 1946 study by Dixon and Yockey,[†] who made a preliminary analysis of rocket propulsion using low-molecular-weight propellants heated in a nuclear reactor. Another was a brief note² by Serber concerned with this concept as well as with direct use of alpha and beta particles, fission frag-

[†] Superscript numbers are keyed to references at the end of the chapter.

ments, etc. Also, at about this time, S. Ulam of the Los Alamos Scientific Laboratory first proposed the propulsion of rocket vehicles by atomic bombs, using the blast pressures generated by their explosion to push on the back end of missiles.

The first comprehensive and detailed study of potentialities and problems of nuclear flight propulsion was made later in 1946 by a group of engineers and physicists at the Applied Physics Laboratory of the John Hopkins University. The secret document³ summarizing this work dealt with the propulsion of nuclear rockets, aircraft, and ramjets. It concluded that hydrogen was the best nuclear-rocket propellant, and graphite and metallic carbides the most likely materials for pile construction for rockets, and that beryllium oxide was a prime contender for use in air-breathing nuclear systems. This study also pointed out the several principles involved and the major problems requiring solution for the use of fission energy in flight-propulsion systems. Foremost among these was the problem of insufficient data on materials properties to allow design work to proceed. It is often still true today, as was written then, that "the necessary data do not exist, and their accumulation is the *sine qua non* for further progress in this field."

This report was followed in 1947 by a more voluminous and more specific secret study⁴ made by the staff of the North American Aviation Corporation, Aerophysics Laboratory. Nuclear-powered ramjets and rockets of several specific sizes for military application were investigated. In the absence of materials data, assumptions were made about the high-temperature strength and corrosion resistance of materials thought to be of interest; these included uranium-loaded graphite and beryllium oxide for rocket and ramjet use, respectively. An important result of this study was the initiation of experimental work to study these properties.

Meanwhile, in May, 1946, the United States Air Force had established the Nuclear Energy for Propulsion of Aircraft (NEPA) project, at Oak Ridge, Tenn., for the purpose of exploring possibilities for nuclear flight. In 1947, secret work on nuclear rockets was begun on a small scale as part of NEPA, and this continued intermittently until 1949. However, NEPA's principal concern was with nuclear aircraft, which were studied extensively from 1946 to 1951, when the project was finally disbanded in favor of a joint AEC/USAF Aircraft Nuclear Propulsion (ANP) program to carry out engineering development of aircraft reactors and engine systems. This action was a partial result of studies made by an *ad hoc* group convened in 1948 at MIT by the AEC to provide an independent assessment of potentialities for nuclear flight. This study group, known as the Lexington Project, came to the conclusion that nuclear aircraft (manned) were likely less difficult than nuclear ramjets, which, in turn, would be less difficult than nuclear rockets to develop—an inversion of the proper order apparent from this vantage point in time. Tsien's brief study⁵ of nuclear rockets using reactors of porous graphite appeared as an outgrowth of this project.

Also in 1948, and early in 1949, Shepherd and Cleaver published a class series of papers⁶ on nuclear-powered rockets. These articles in the open literature surveyed in thorough fashion the general methods of application of fission energy to rocket propulsion and introduced new concepts such as the gaseous-core reactor. Their papers discuss nearly all the basic forms of nuclear-rocket systems considered since that time.

Interest in nuclear rockets then seemed to die almost completely until a detailed engineering study⁷ was made at the Oak Ridge National Laboratory, late in 1952, of the potentialities of nuclear heat-exchanger rockets for a wide range of mission applications, based upon the then known state of reactor technology. Emphasis was placed upon the use of graphite fuel-element structures, and the study was parameterized to cover a wide range of missions in order to identify regions of desirable performance.

Following this, attention in the United States returned once again to the great advances possible by nuclear rockets, and more theoretical studies of performance were undertaken, at the request of the United States Air Force which was interested principally in military applications. Among other things, the Lawrence Radiation Laboratory (Livermore) and the Los Alamos Scientific Laboratory were asked to look at the problem at this time and report their conclusions to the Nuclear Missiles subcommittee of the USAF Scientific Advisory Board. Early in 1955 Los Alamos summarized its investigations in a secret report⁸ concerning nuclear-powered ballistic missiles. The performance advantages shown for nuclear rockets led to support for materials research in the field and to the initiation of serious nuclear-rocket reactor work at both laboratories. Activity continued vigorously for about a year, at which time the Livermore Laboratory was redirected to work on nuclear ramjets, the Pluto program, whereas Los Alamos continued with nuclear-rocket development which has led to the current Rover program. The first nuclear-rocket reactor test (Kiwi-A) was carried out in July, 1959, and the first nuclear-ramjet reactor (Tory-II) was operated in spring, 1961, at the Nevada Test Site of the AEC. These are shown in Figs. 1-1 and 1-2, respectively.

Meanwhile nuclear-aircraft work proceeded apace following establishment of the joint AEC/USAF development program in 1951, but did not fare as well as the later rocket or ramjet programs. ANP took two roads from among the great diversity and complexity of technical approaches:† on the route of the direct cycle in which air was heated by passage through reactor; the other indirect, transferring reactor heat to air from heated fluid (liquid metal) in a separate heat exchanger outside the reactor. Plagued with continuing reviews and changes of requirements as well as with numerous real technical problems, neither course had yielded a successful development of flight prototype engines by the time the ANP program was

† An excellent discussion and summary of the many possible engine cycles for nuclear aircraft was given in 1957 in a Russian publication⁹ on nuclear-flight-propulsion systems.

canceled, after ten years, in 1961. However, much of the nuclear-reactor technology gained in this program may yet bear successful fruit in application to flight-propulsion and space power systems.

Several points stand out from this chronology. First, we see that the order of ten years of development time is required to assess the ability of engineering technology to achieve true potentialities of a complex nuclear-flight-propulsion system, e.g., ANP. The basic reason that the ANP

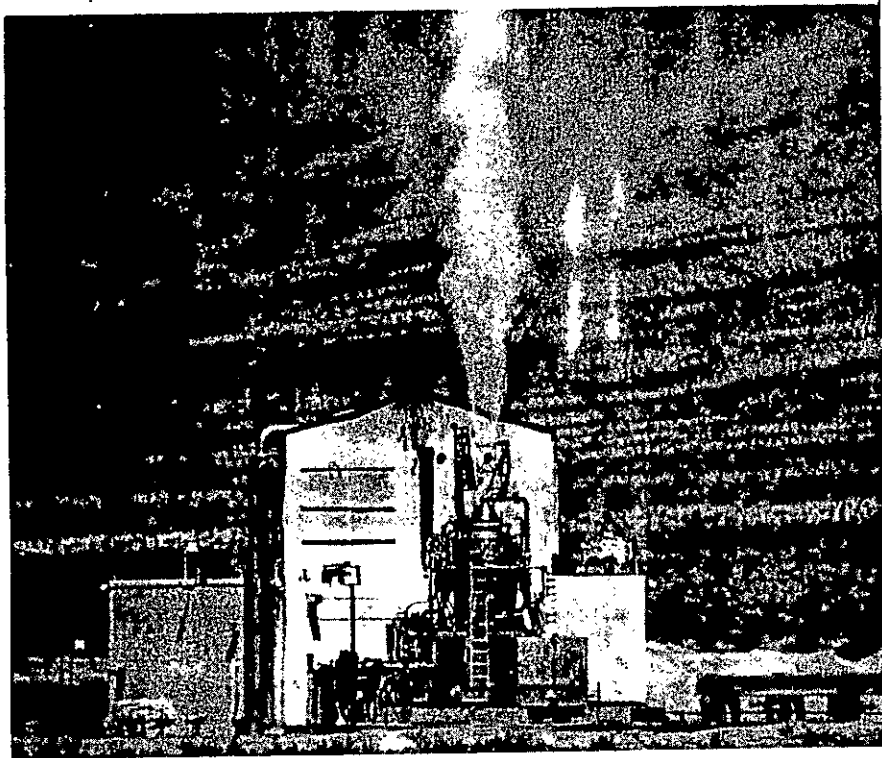


Fig. 1-1 First test of a nuclear-rocket reactor: Kiwi-A reactor operating at full power, NTS, July 1, 1959. (Photo courtesy of Los Alamos Scientific Laboratory, Los Alamos, N.Mex.)

program was not continued to conclusion was the fact that this ten-year assessment did not provide assurance of meeting requirements thought desirable for advanced manned vehicles. Far from being the simplest of the three—nuclear aircraft, rockets, and ramjets—R. C. Briant, the director of the ANP Project at Oak Ridge National Laboratory, held in 1954 that “manned nuclear aircraft pose the most difficult engineering development job yet attempted within this century.” The job was simply too hard for the then available or attainable technology.

This brings us to the second point. During this period power-reactor development was under way in many places, both in the United States and

abroad. This too had its setbacks, but has led to a practical nuclear power industry. Why the success in one area and not the other? The answer is simple: vastly different requirements on system mass per unit power output. There is very little incentive to minimize the mass of ground- or ship-based power plants, and their design features are chosen with regard to other problems, chiefly economic, such as ease of maintenance, simplification of fuel processing, and cheapness of construction. In contrast, for nuclear flight propulsion, considerations such as these must play a secondary role to the overriding need to minimize the mass/power ratio, consistent with

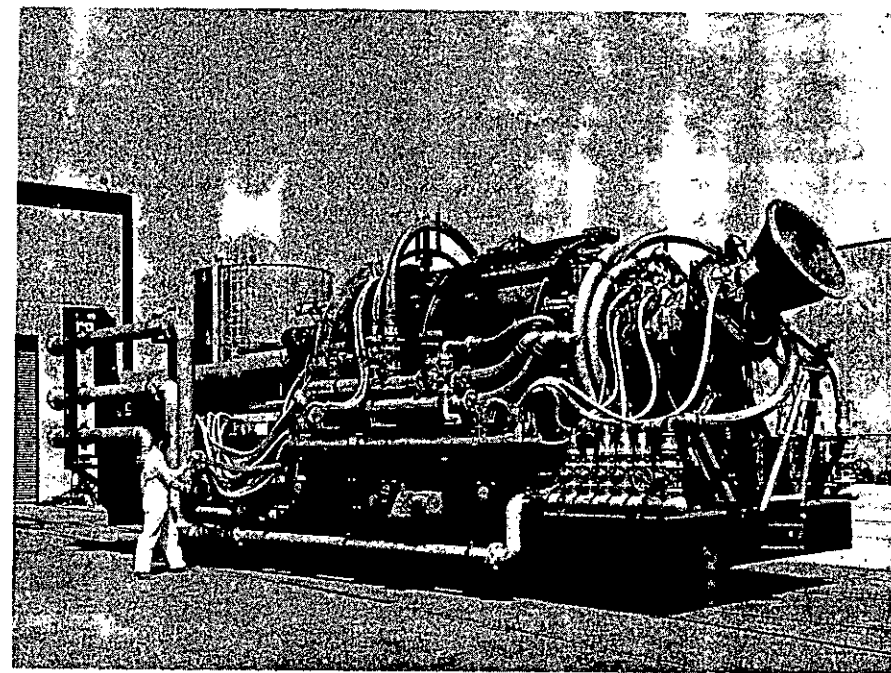


Fig. 1-2 The first nuclear-ramjet reactor: Tory-IIa reactor tested at NTS, May, 1961. (Photo courtesy of Lawrence Radiation Laboratory, Livermore, Calif.)

requirements of system reliability. High performance needs greater knowledge, costs more, and is more difficult to attain than is low performance. Unless commanding advantages in performance and eventual concurrent real benefits in overall system use can be shown to be attainable, nuclear-flight-propulsion systems will not be adopted.

The requirements of space flight are so demanding that they stretch chemical-rocket technology to the utmost. For example, a multiple-stage chemical-rocket system as massive as a small naval cruiser is presently required to send two or three men to the Moon with safe return. Such a course is reasonable only if the job must be done and there is no other way